OPLOGS-07: Analytics & Dashboards

> **Series:** OPLOGS — OpenPipeline Logs | **Notebook:** 7 of 8 | **Created:** December 2025 | **Last Updated:** 04/25/2026

Aggregation, Time Series, and Visualization Queries

This notebook covers aggregation functions, time series analysis, statistical patterns, and dashboard-ready queries for log analytics.

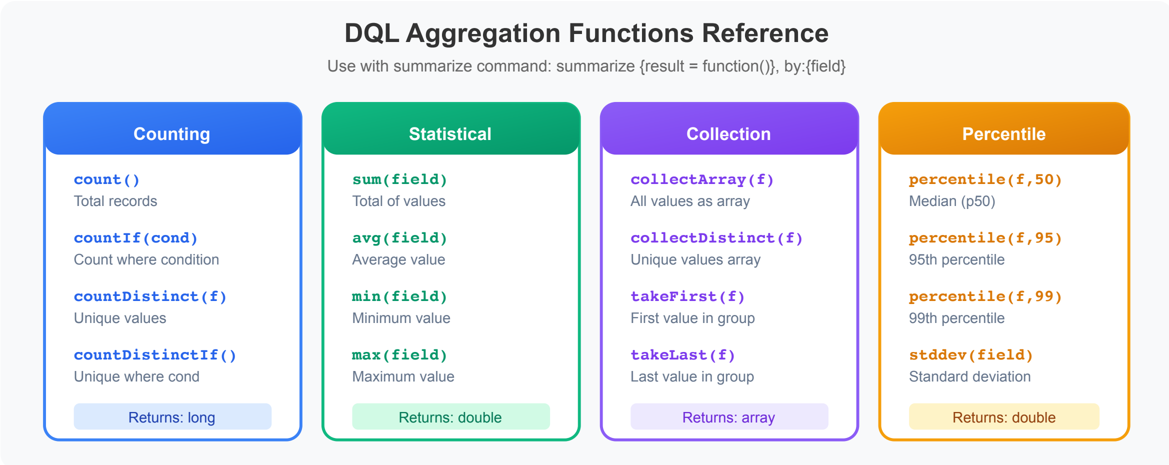
Table of Contents

Prerequisites

- ✅ Access to a Dynatrace environment with log data
- ✅ Completed OPLOGS-01 through OPLOGS-06

- ✅ Familiarity with dashboard concepts

1. Aggregation Functions



Common Aggregation Functions

Function	Description	Example
`count()`	Count records	`summarize {total = count()}`
`countIf(condition)`	Conditional count	`countIf(loglevel == "ERROR")`
`countDistinct(field)`	Unique values	`countDistinct(k8s.pod.name)`
`sum(field)`	Sum values	`sum(bytes)`
`avg(field)`	Average	`avg(response_time)`
`min(field)`	Minimum	`min(timestamp)`
`max(field)`	Maximum	`max(timestamp)`
`takeFirst(field)`	First value	`takeFirst(content)`
`takeLast(field)`	Last value	`takeLast(content)`
`collectDistinct(field)`	Array of unique values	`collectDistinct(loglevel)`

🚨 Important: Named Aliases Required

```
// ✅ CORRECT - Named aliases
| summarize {count = count(), errors = countIf(...)}, by: {...}
| sort count desc

// ❌ WRONG - Anonymous aggregation can't be used in sort
| summarize count(), by: {...}
| sort count() desc  // ERROR!
```

```
// Basic aggregations
fetch logs, from: now() - 1h
| summarize {
    total_logs = count(),
    unique_hosts = countDistinct(dt.entity.host),
    unique_namespaces = countDistinct(k8s.namespace.name),
    unique_pods = countDistinct(k8s.pod.name)
  }
```

```
// Aggregations with grouping
fetch logs, from: now() - 1h
| summarize {
    total = count(),
    errors = countIf(loglevel == "ERROR"),
    warnings = countIf(loglevel == "WARN"),
    info = countIf(loglevel == "INFO")
  }, by: {k8s.namespace.name}
| sort total desc
| limit 15
```

```
// Error rate calculation
fetch logs, from: now() - 1h
| summarize {
    total = count(),
    errors = countIf(loglevel == "ERROR")
  }, by: {k8s.namespace.name}
| filter total &gt; 100  // Minimum sample size
| fieldsAdd error_rate_pct = round((errors * 100.0) / total, decimals: 2)
| sort error_rate_pct desc
| limit 15
```

2. Time Series Analysis

The `makeTimeseries` command creates time-bucketed data for trend visualization.

```
// Log volume over time (5-minute buckets)
fetch logs, from: now() - 6h
| makeTimeseries {log_count = count()}, interval: 5m
```

```
// Error trend over time
fetch logs, from: now() - 6h
| makeTimeseries {
    total = count(),
    errors = countIf(loglevel == "ERROR"),
    warnings = countIf(loglevel == "WARN")
  }, interval: 5m
```

```
// Time series by dimension (namespace)
fetch logs, from: now() - 6h
| filter isNotNull(k8s.namespace.name)
| makeTimeseries {
    log_count = count()
  }, by: {k8s.namespace.name}, interval: 10m
```

```
// Error rate time series by host
fetch logs, from: now() - 6h
| filter isNotNull(dt.entity.host)
| makeTimeseries {
    total = count(),
    errors = countIf(loglevel == "ERROR")
  }, by: {dt.entity.host}, interval: 15m
```

3. Statistical Analysis

```
// Log volume statistics by source
fetch logs, from: now() - 24h
| summarize {
    total = count(),
    first_seen = min(timestamp),
    last_seen = max(timestamp)
  }, by: {dt.openpipeline.source}
| fieldsAdd duration_ns = toLong(last_seen) - toLong(first_seen)
| fieldsAdd duration_hours = round(duration_ns / 3600000000000.0, decimals:
1)
| fieldsAdd logs_per_hour = round(total / (duration_ns / 3600000000000.0),
decimals: 0)
| sort total desc
```

```
// Percentile analysis (if numeric field available)
fetch logs, from: now() - 1h
| fieldsAdd content_length = stringLength(content)
| summarize {
    avg_length = avg(content_length),
    min_length = min(content_length),
    max_length = max(content_length),
    total_logs = count()
  }, by: {k8s.namespace.name}
| sort avg_length desc
| limit 15
```

```
// Hourly distribution analysis
fetch logs, from: now() - 24h
| fieldsAdd hour_bucket = bin(timestamp, 1h)
| summarize {log_count = count()}, by: {hour_bucket}
| sort hour_bucket asc
```

```
// Daily distribution analysis
fetch logs, from: now() - 7d
| fieldsAdd day_bucket = bin(timestamp, 1d)
| summarize {
    log_count = count(),
    error_count = countIf(loglevel == "ERROR")
  }, by: {day_bucket}
| sort day_bucket asc
```

4. Dashboard-Ready Queries

These queries are optimized for dashboard tiles and visualizations.

```
// Single Value: Total Logs (Last Hour)
fetch logs, from: now() - 1h
| summarize {total_logs = count()}
```

```
// Single Value: Error Count (Last Hour)
fetch logs, from: now() - 1h
| filter loglevel == "ERROR"
| summarize {error_count = count()}
```

```
// Pie Chart: Log Level Distribution
fetch logs, from: now() - 1h
| summarize {count = count()}, by: {loglevel}
| sort count desc
```

```
// Bar Chart: Top Error Sources
fetch logs, from: now() - 1h
| filter loglevel == "ERROR"
| summarize {error_count = count()}, by: {k8s.namespace.name}
| sort error_count desc
| limit 10
```

```
// Line Chart: Log Trend (6 Hours)
fetch logs, from: now() - 6h
| makeTimeseries {
    errors = countIf(loglevel == "ERROR"),
    warnings = countIf(loglevel == "WARN"),
    info = countIf(loglevel == "INFO")
  }, interval: 5m
```

```
// Table: Top Error Messages
fetch logs, from: now() - 1h
| filter loglevel == "ERROR"
| fieldsAdd error_preview = substring(content, from: 0, to: 100)
| summarize {
    occurrences = count(),
    first_seen = min(timestamp),
    last_seen = max(timestamp)
  }, by: {error_preview, k8s.namespace.name}
| sort occurrences desc
| limit 20
```

```
// Heat Map: Errors by Hour and Namespace
fetch logs, from: now() - 24h
| filter loglevel == "ERROR"
| fieldsAdd hour_bucket = bin(timestamp, 1h)
| summarize {error_count = count()}, by: {hour_bucket, k8s.namespace.name}
| sort hour_bucket asc
```

5. Trend Analysis

```
// Compare current hour to previous hour
fetch logs, from: now() - 2h
| fieldsAdd period = if(timestamp &gt; now() - 1h, "current", else:
"previous")
| summarize {count = count()}, by: {period, loglevel}
| sort period asc, count desc
```

```
// New error patterns (appeared in last hour)
fetch logs, from: now() - 1h
| filter loglevel == "ERROR"
| fieldsAdd error_sig = substring(content, from: 0, to: 80)
| summarize {
    count = count(),
    first_seen = min(timestamp)
  }, by: {error_sig, k8s.namespace.name}
| filter first_seen &gt; now() - 30m  // First seen in last 30 minutes
| sort first_seen desc
| limit 15
```

```
// Volume anomaly detection (compare to baseline)
fetch logs, from: now() - 6h
| fieldsAdd time_bucket = bin(timestamp, 15m)
| summarize {log_count = count()}, by: {time_bucket, k8s.namespace.name}
| sort time_bucket desc
| limit 100
```

6. Log Pattern Analysis

```
// Top log patterns (by content prefix)
fetch logs, from: now() - 1h
| fieldsAdd pattern = substring(content, from: 0, to: 60)
| summarize {count = count()}, by: {pattern}
| sort count desc
| limit 25
```

```
// Unique log levels and statuses
fetch logs, from: now() - 1h
| summarize {
    loglevels = collectDistinct(loglevel),
    statuses = collectDistinct(status),
    sources = collectDistinct(dt.openpipeline.source)
  }
```

```
// Log diversity score (unique patterns per namespace)
fetch logs, from: now() - 1h
| fieldsAdd pattern = substring(content, from: 0, to: 50)
| summarize {
    total_logs = count(),
    unique_patterns = countDistinct(pattern)
  }, by: {k8s.namespace.name}
| fieldsAdd diversity_ratio = round((unique_patterns * 100.0) / total_logs,
decimals: 2)
| sort diversity_ratio desc
| limit 15
```

7. Operational Dashboard Queries

```
// Operations Summary (last 1h)
fetch logs, from: now() - 1h
| summarize {
    total_logs = count(),
    error_count = countIf(loglevel == "ERROR"),
    warn_count = countIf(loglevel == "WARN"),
    unique_hosts = countDistinct(dt.entity.host),
    unique_pods = countDistinct(k8s.pod.name)
  }
| fieldsAdd error_rate = round((error_count * 100.0) / total_logs, decimals: 2)
```

```
// Health scorecard by namespace
fetch logs, from: now() - 1h
| filter isNotNull(k8s.namespace.name)
| summarize {
    total = count(),
    errors = countIf(loglevel == "ERROR"),
    warnings = countIf(loglevel == "WARN")
  }, by: {k8s.namespace.name}
| fieldsAdd error_rate = round((errors * 100.0) / total, decimals: 2)
| fieldsAdd health_status = if(error_rate &gt; 10, "CRITICAL",
                            else: if(error_rate &gt; 5, "WARNING",
                            else: "HEALTHY"))
| sort error_rate desc
```

```
// Ingestion pipeline summary
fetch logs, from: now() - 1h
| summarize {
    log_count = count(),
    unique_pipelines = countDistinct(dt.openpipeline.pipelines)
  }, by: {dt.openpipeline.source, dt.system.bucket}
| sort log_count desc
```

📝 Summary

In this notebook, you learned:

✅ **Aggregation functions** - count, countIf, countDistinct, sum, avg
✅ **Time series** - makeTimeseries with intervals
✅ **Statistical analysis** - hourly/daily patterns, percentiles
✅ **Dashboard queries** - Single values, charts, tables
✅ **Trend analysis** - Period comparison, anomaly detection
✅ **Pattern analysis** - Log clustering and diversity

➡️ Next Steps

Continue to **OPLOGS-08: Security & Data Protection** for data protection patterns.

📚 References

- DQL Aggregation Functions
- DQL makeTimeseries
- Dynatrace Dashboards
